Water Supply and Wastewater Engineering

CEP Report

WATER SUPPLY SYSTEM

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1 Executive Summary

This report explores the design of a water supply system for Muslim Nagar Housing Society, Lahore, to meet the growing needs of its residents over the next 20 years. The society, spread over 18 hectares, currently relies on individual tube wells and water tankers, which are costly and inefficient. Two solutions were analyzed: a direct pumping system and a system with an overhead reservoir (OHR).

The direct pumping system delivers water directly into the network using pumps, making it simpler and more affordable, with a total cost of PKR 16.49 million over 20 years. However, it depends entirely on electricity, which could disrupt the supply during power outages. On the other hand, the OHR system distributes water using gravity, ensuring steady pressure and reliability, even during outages. While more robust, it comes with a higher total cost of PKR 24.79 million over 20 years.

After careful analysis, the direct pumping system is recommended as the more economical and practical choice, with a monthly household cost of PKR 139 compared to PKR 211 for the OHR system. However, if the need for greater reliability or concerns about power outages arise, the OHR system could be reconsidered in the future. This solution strikes a balance between cost, efficiency, and sustainability to meet the society's water needs effectively.

Table of Contents

1 Executive Summary 2
2 Background 8
3 Project Description 8
3.1 Facilities 9
4 DESIGN CONSIDERATION 9
4.1 Design Criteria 9
4.1.1 Pressure 9
4.1.2 Demand 9
4.1.3 Velocity in Pipes 9
4.1.4 Unit Head-loss 9
4.1.5 Cover over Pipes 9
4.1.6 Fire Demand and Fire Hydrants Locations 9
4.1.7 Minimum Pipe Sizes 10
5 General Information 10
5.1 Sources of Water 10
5.2 Components of Water Supply System 10
5.3 Pressure in Water Distribution System 10
5.4 Design Period & Population Estimation 11
5.5 Water Demands 11
5.6 Factors Affecting Water Consumption 11
5.7 Types of Distribution System (Based on Pumping) 11
There are three types of Distribution system based on pumping: 11
5.8 Types of Supply 11
5.9 Types of System (Based on Design) 11
6 DESIGN OF WATER SUPPLY SYSTEM 12
6.1 Population Projection 12
6.2 Design Option 1: Direct Pumping 12
6.2.1 Water Demand Calculations: 12
6.2.2 Design of Tube Well 13
6.2.3 Pipes 14
6.2.4 Valves: 15
6.2.5 Distribution Network 15
6.2.6 Maintenance Considerations 15
6.3 Design Option 2: Pumping with Storage 16

6.3.1 Water demand Calculation 16
6.3.2 Overhead Reservoir (OHR) Design 16
6.3.3 Tube Well Design 16
6.3.4 Motor BHP Calculation 17
6.3.5 Pipe Optimization 18
7 Complete Design of Distribution System (Using WaterGEMS) 18
7.1 Model Setup 18
7.2 Nodes and Demand Allocation: 19
7.3 Pipes: 19
7.4 Pump and Reservoir 19
7.4.1 For Direct Pumping: 19
7.4.2 For Pumping with OHR: 19
7.5 Simulation Results (WaterGEMS Output) 19
7.5.1 Hydraulic Head at Nodes: 19
7.5.2 Pipe Velocities: 19
7.5.3 Energy Requirements: 19
8 Pipe Summaries (Using WaterGEMS Reports) 19
8.1 Valve Placement 20
8.2 Attachments from WaterGEMS 20
9 Cost Invloved 20
9.1 Pipe Costs 20
9.2 Valve Costs 20
9.2.1 Sluice Valves: 20
9.2.2 Air Release Valves: 21
9.2.3 Check Valves: 21
9.3 Excavation 21
9.4 Pumping System 21
9.5 Overhead Reservoir 21
9.6 Total Capital Costs: 22
9.7 Electricity Costs 22
9.8 Maintenance Costs 23
9.9 Total Operational Costs (20 Years): 23
9.10 Total Cost Over 20 Years 23
10 Staffing Levels and Costs 24
10.1 Staffing Levels 24

10.2 Cost of Staffing 24
10.3 Justification for Staffing 25
11 Final Comparison and Recommendation for Muslim Nagar Society 25
11.1 Direct Pumping: 25
11.2 Pumping with OHR: 25
11.3 Recommendation for Muslim Nagar Society: 25
12 Monthly Household Bills for Operational Costs 26
12.1 Total Operational Cost (Monthly): 26
12.2 Monthly Bills 26
12.3 Summary: 26
12.4 Key Observations 26
12.5 Recommendation: 27
13 References 28
14 Annexure 29

List of Tables

Table 1. Pressure pipe inventory Table 20
Table 2. Total Cost of all Components 22
Table 3. Total Maintenance cost....23
Table 4. Total cost over 20 years 23
Table 5. Staff levels 24
Table 6. Staff Costing 24
Table 7. Total Cost Summary 26

List of Figures

Figure 1. Map of Muslim Nagar Housing Society.... 8
Figure 2. GI Pipes....14
Figure 3. Sluice Valve....15
Figure 4. Complete Direct Pumping System....15
Figure 5. Complete Pumping with storage system 18

Abbreviations Used

1. **Psi** - Pounds per square inch
2. **LPCD** - Liters per capita per day
3. **GIS** - Geographic Information System
4. **CAD** - Computer-Aided Design
5. **GPM** - Gallons per minute
6. **GI** - Galvanized Iron
7. **OHR** - Overhead Reservoir
8. **PHED** - Public Health Engineering Department
9. **MRS** - Market Rate Schedule
10. **RCC** - Reinforced Cement Concrete
11. **HGL** - Hydraulic Grade Line
12. **UFW** - Unaccounted-for Water
13. **BHP** - Brake Horsepower
14. **HP** - Horsepower
15. **ft/sec** - Feet per second
16. **cfs** - Cubic feet per second
17. **PVC** - Polyvinyl Chloride
18. **FOS** - Factor of Safety
19. **PKR** - Pakistani Rupee

2 Background

Muslim Nagar Housing Society, situated along Jati Umrah Road, Lahore, is a residential area experiencing rapid urbanization. With an increasing population and rising water demand, the need for a reliable and sustainable water supply system is critical. Currently, no centralized water supply infrastructure exists, compelling the residents to rely on individual tube wells and water tankers. This results in inefficiencies, uneven water quality, and increased costs.

This project aims to design a comprehensive water supply system tailored to the unique requirements of the society. The design will consider technical, environmental, and socio-economic factors to ensure optimal functionality for the next 20 years.



Figure 1. Map of Muslim Nagar Housing Society

3 Project Description

Muslim Nagar Housing Society is a well-planned residential area located along Jati Umrah Road, Lahore, Pakistan. The society spans approximately **18 hectares** *(Google Earth)* and serves middle and upper-middle-class residents. Key details of the project area include

Coordinates that are approximately **31.5204° N, 74.3587° E** *(Literature)*, with easy access via the main Lahore Ring Road. Lahore experiences hot summers (up to **45°C**), mild winters (as low as **5°C**), and receives an average annual rainfall of **~625 mm**, predominantly during the monsoon season.

3.1 Facilities

Residential Plots are about **416 residential plots** of varying sizes*. (Society head office)*

Community Amenities includes **one school**, **one mosque**, **five parks**, and two **commercial** establishments.

This report aims to design a sustainable and efficient water supply system to meet the growing needs of the society.

4 DESIGN CONSIDERATION

The whole design of Water Supply System follows the standard design considerations based on the criteria adopted from the Water and Sanitation Agency (WASA) and the Public Health Engineering Department (PHED):

4.1 Design Criteria

4.1.1 Pressure

- **Minimum residual pressure:** 20 psi at the consumer's end.
- **Maximum pressure:** 80 psi to avoid bursting of pipes and fittings.

4.1.2 Demand

- **Residential Areas**: Average daily water consumption per capita: 45 LPCD (liters per capita per day) with a 15% addition for Unaccounted-for Water (UFW).
- **Commercial Areas**: Water consumption based on specific needs such as hotels, offices, and shops.
- **Parks and Public Spaces:** Additional water for irrigation based on area and type of vegetation.
- **Mosques:** Additional water usage for ablution and sanitation.

4.1.3 Velocity in Pipes

- **Recommended velocity**: 1-5 ft/sec to ensure efficient flow and minimize sedimentation.

4.1.4 Unit Head-loss

- Maximum allowable head loss: 10 ft per 1000 ft of pipe length to maintain adequate pressure.

4.1.5 Cover over Pipes

- **Minimum cover:** 3 feet to protect pipes from external loads and prevent freezing in colder areas.

4.1.6 Fire Demand and Fire Hydrants Locations

- **Fire demand**: 1500 liters per minute for a duration of 2 hours.
- Fire hydrants should be located within 300 ft of each other and near high-risk areas

like schools, hospitals, and commercial zones.

4.1.7 Minimum Pipe Sizes

- **Minimum diameter for mains:** 6 inches to ensure sufficient flow and pressure.
- **Service connections**: ¾ inch to 1 inch based on household requirements.

These criteria ensure a reliable and efficient water supply system, maintaining adequate pressure, meeting varying demands, and allowing for effective firefighting capabilities

5 General Information

5.1 Sources of Water

Water sources can be categorized into **surface water, groundwater, and rainwater.** Surface water includes rivers, lakes, and reservoirs, which collect water from precipitation and runoff. Groundwater, stored in aquifers beneath the earth's surface, is extracted through wells and tube wells. Additionally, rainwater harvesting systems are used to collect and store rainwater, reducing dependence on other water sources. The main water source of the society is the **Ground Water.**

5.2 Components of Water Supply System

- **Overhead Tanks (OHT):** Store water at height to provide pressure to the distribution system.
- **Reservoirs:** Large storage tanks or natural basins that hold water for supply.
- **Pipes**: Transport water from the source to consumers. Materials include PVC, steel, and concrete.
- **Valves**: Regulate water flow, pressure, and direction within the system.
- **Pumps:** Lift or move water to higher elevations or over long distances.
- **Fire Hydrants**: Provide access to water for firefighting.
- **Fittings:** Connect pipes and other system components.
- **Valve Chambers:** House valves to protect them from damage and facilitate maintenance.
- **Transmission Mains:** Large pipes that carry water from the source to the distribution network.
- **Distribution Mains:** Smaller pipes that deliver water to individual consumers.

5.3 Pressure in Water Distribution System

Pressure in a water distribution system can be classified into **static, residual, and working pressure.** Static pressure refers to the pressure in the system when water is **not** in motion. Residual pressure is the remaining pressure while water is being consumed. Working pressure is the level of pressure necessary for the efficient operation of water fixtures and appliances.

5.4 Design Period & Population Estimation

- **Design Period:** The timeframe for which the water supply system is planned, typically spanning 15 to 30 years *(PHED design criteria)* and here 20 years is taken.
- **Population Estimation:** Calculating future population based on current data and growth rates to ensure the system can handle future demands.

5.5 Water Demands

- **Average Daily Demand:** The mean amount of water used per day by the community.
- **Maximum Daily Demand**: The highest daily water usage, usually calculated as 1.5 times the average daily demand.
- **Peak Hourly Demand:** The highest hourly water usage within a day, used to size pumps and other components.

5.6 Factors Affecting Water Consumption

- **Climate**: Hot climates increase water use for cooling, irrigation, etc.
- **Population Density**: Higher densities mean more water usage per area.
- **Lifestyle**: Different socio-economic conditions and living standards affect water usage.
- **Industrial Activity**: Factories and industries consume significant amounts of water.
- **System Losses:** Water lost due to leaks, theft, or other inefficiencies.

5.7 Types of Distribution System (Based on Pumping)

There are three types of Distribution system based on pumping:

- **Direct Pumping:** Water is pumped directly into the distribution system without intermediate storage.
- **Gravity System**: Water stored in elevated tanks and distributed by gravity.
- **Combined System:** Uses both direct pumping and gravity-fed storage for distribution.

5.8 Types of Supply

- **Continuous Supply**: Water is available 24/7 without interruption.
- **Intermittent Supply**: Water is supplied only during certain times of the day.

In Muslim Nagar, Continuous supply of water is provided.

5.9 Types of System (Based on Design)

- **Loop System**: Network forms loops, allowing water to flow in multiple directions, improving reliability and pressure.
- **Dead End System**: Network has terminal points (dead ends), which can lead to water stagnation and pressure issues.

6 DESIGN OF WATER SUPPLY SYSTEM

6.1 Population Projection

The society has overall **416** plots, among which **350** are constructed with residents residing in them. 300 of 350 are double storey and 50 are single storey houses.

Assuming a Double storey house has 14 people living and a single storey house has 7 people,

Total Population = (50 * 7) + (300 * 14) = 4550 people

Assuming a **2.5% annual growth rate of Lahore**, the projected population for the next **20 years** is calculated as follows:

$$Pfuture = Pcurrent * (1 + growth\ rate)\text{^}years$$

Substituting the values:

$$Pfuture = 4550 * (1 + 0.025)^{20} = 7456\ people$$

Projected Population (20 years): **7,456 people**

This figure will be used to design the water supply system.

6.2 Design Option 1: Direct Pumping

In this system, water is directly pumped from the tube well into the distribution network without intermediate storage. This requires pumps that can sustain **peak hourly flow rates** while maintaining residual pressure in the network.

6.2.1 Water Demand Calculations:

Per Capita Water Consumption: **45 LPCD (**PHED Punjab design criteria).

Adding **15% for (UFW)** (e.g., losses due to leaks and inefficiencies), the total water demand is calculated as:

$$Total\ daily\ water\ demand = Pfuture * Per\ capita\ consumption * (1 + UFW)$$

$$Total\ daily\ water\ demand = 7456 * 45 * (1 + 0.15) = 385{,}848\ litres/day = 385.8\ m3/day$$

Average Daily Demand: $385.8\ m3/day$

Peak Hourly Demand:

$$Peak\ hourly\ demand = 2.25 * Avg.daily\ demand = 868\ m3/day = 36.23\ m3/hr$$

Residual Pressure

Minimum residual pressure in the network is **12 m (40 ft)**.

6.2.2 Design of Tube Well

To meet the daily and peak demands, the tube well(s) must be designed accordingly. Using PHED Punjab design criteria:

6.2.2.1 Discharge of Tube Well:

Tube well discharge is calculated based on the peak demand:

$$Discharge = \frac{Peak\ Demand}{Operational\ hours}$$

Assuming the tube well operates for **18 hours/day**:

$$Discharge = \frac{868}{18}$$

= 48.2 m^3/hr

The pump must sustain **peak hourly flow rates** with sufficient head to overcome:

$$Total\ Head = Static\ Head + Dynamic\ Head$$

Static Head:

- Minimum residual pressure: **40 ft**
- Total Static Head: 40ft

Dynamic Head:

$$Dynamic\ Head = Head\ losses\ in\ pipe + chamber\ losses + FOS$$

Head losses in pipe is found out by using the **Hazen-Williams Equation** (FPS system):

$$Hf = 4.73 * \left(\frac{Q}{C}\right)^{1.85} * L/D^{4.87}$$

Where:

L=266 ft (total length of the pipe for direct pumping)

C=140 (Hazen-Williams coefficient for PVC)

d=0.5 ft (pipe diameter, 6 inches converted to feet)

Q=48.2 m^3/hour = 0.473 cfs (peak hourly flow rate)

Substituting,

= 0.986 ft + 10 % for specials in rising main

= 1.08 ft

$$Dynamic\ Head = Head\ losses\ in\ pipe + chamber\ losses + FOS$$

Chamber losses = 5 ft

FOS = 10 ft

Total Pump Head:

$$Total\ Head = 40 + 1.08 + 5 + 10$$

= 56.1 ft

6.2.2.2 Motor Sizing:

Using the formula:

$$BHP = \frac{\gamma \cdot Q \cdot H}{550 * \eta}$$

Substituting:

$$BHP = \frac{62.4 * 0.473 * 56.1}{550 * 0.6}$$

= 5.02

A **10 HP motor** is recommended.

6.2.3 Pipes



Figure 2. GI Pipes

GI pipes: Chosen for their durability and resistance to corrosion.

Pipe Diameter in Rising Main: 6 inches (velocity: **1.61 ft/sec**, within limits of 1.5–7 ft/sec).

6.2.4 Valves:



Figure 3. Sluice Valve

- **Sluice Valves**: Installed at the pump outlet to isolate the pump.
- **Check Valves**: Prevent backflow from the distribution system to the pump.
- **Air Release Valves**: Placed at high points along the main to release trapped air.

6.2.5 Distribution Network

1. **Layout**: Grid-iron system.

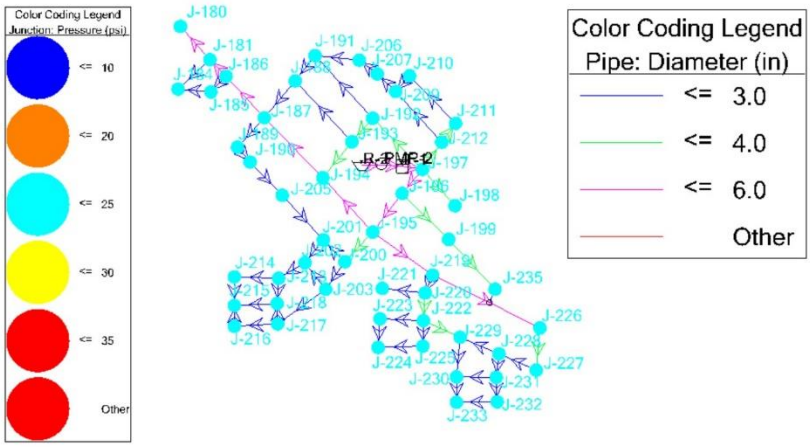


Figure 4. Complete Direct Pumping System

2. **Pipe Diameters**: Minimum of 3 inches (75 mm) for distribution pipes.
3. **Velocity Check**: Maintained between 1–5 ft/sec.
4. **Valves**:

- **Sluice Valves**: For isolating pipe sections.
- **Pressure Reducing Valves**: Used where excess pressure is observed.

6.2.6 Maintenance Considerations

- Direct pumping systems require more frequent pump maintenance due to continuous operation.
- Dependency on electricity increases operational costs.
- Pressure fluctuations during peak demand may affect network reliability.

6.3 Design Option 2: Pumping with Storage

6.3.1 Water demand Calculation

Avg Day Demand: $385.8\ m3/day$

Peak Hourly Demand: $36.23\ m3/hr$

Maximum Day Demand**:**

$$Max\ Day\ Demand = Avg\ Daily\ Demand * 1.5$$

= 385.8 * 1.5

≈ 580 m^3/day

6.3.2 Overhead Reservoir (OHR) Design

According to PHED Punjab design criteria, the capacity of the OHR should be **1/10th of the Average Daily Demand**:

$$OHR\ capacity = \frac{Avg\ Daily\ Demand}{10}$$

$$OHR\ capacity = \frac{385.8}{10} = \mathbf{38.65\ m3/day}$$

Proposed OHR Capacity: **39 m^3 (rounded).**

6.3.3 Tube Well Design

For systems with an OHR, the tube well discharge is based on the **Maximum Day Demand**:

$$Tubewell\ discharge = \frac{Max\ Daily\ Demand}{Operatinal\ hours}$$

Assuming the tube well operates for **18 hours/day**:

$$Tubewell\ discharge = \frac{580}{18}$$

≈ 32 m^3/hr

The total pump head includes:

Static Head:

Depth of water table: **100 ft**

Drawdown: **20 ft**

Height of OHR: **62 ft**

Elevation difference = 0

$$Total\ Static\ Head = 100 + 20 + 62 = 182\ ft$$

Dynamic Head:

Head losses in pipe are found by using the **Hazen-Williams Equation (FPS)**:

$$Hf = 4.73 * \left(\frac{Q}{C}\right)^{1.85} * L/D^{4.87}$$

Where:

- L=135 ft (length of rising main)
- C =130
- D = 6 in = 0.5 ft
- Q = 32.2 m^3/hour = 0.316 cfs

Substituting values:

Hf = 0.23 ft + 10 % for specials in rising main

Hf = 0.25 ft

Total Pump Head:

Chamber losses = 5 ft

FOS = 10 ft

$$Total\ Pump\ Head = 182 + 0.25 + 5 + 10 = 197.25\ ft$$

- **Proposed Pump Head ≈ 197 ft**

6.3.4 Motor BHP Calculation

The motor brake horsepower (BHP) is calculated as:

$$BHP = \frac{\gamma \cdot Q \cdot H}{550 * \eta}$$

Where:

- Γ = 62.4 lb/ft^3
- Q = 0.316 cfs
- H = 197 ft
- η=0.6 (efficiency)

Substituting:

$$BHP = \frac{62.4 * 0.316 * 197}{550 * 0.6}$$

$$= 11.77$$

- **Proposed Motor Size**: **15 HP** (rounded up).

6.3.5 Pipe Optimization

The following conditions are to be check for pipe optimization.

6.3.5.1 Flow Velocity Check

$$V = \frac{Q}{A} \qquad (A = \pi \times r2)$$

For a 6-inch pipe:

r = 0.5/2 = 0.25 ft

$A=\pi \times (0.25)^2 = 0.196\ ft^2$

$$V = \frac{0.316}{0.196} = 1.61\ ft/sec$$

The velocity is within the PHED criteria of **1–5 ft/sec**.

7 Complete Design of Distribution System (Using WaterGEMS)

The water distribution system of **Muslim Nagar Housing Society** was modeled using **WaterGEMS**. The following steps were performed in the software:

7.1 Model Setup

Network Layout: The society's layout was digitized, and pipelines were placed according to the grid-iron system as shown in Figure 5.

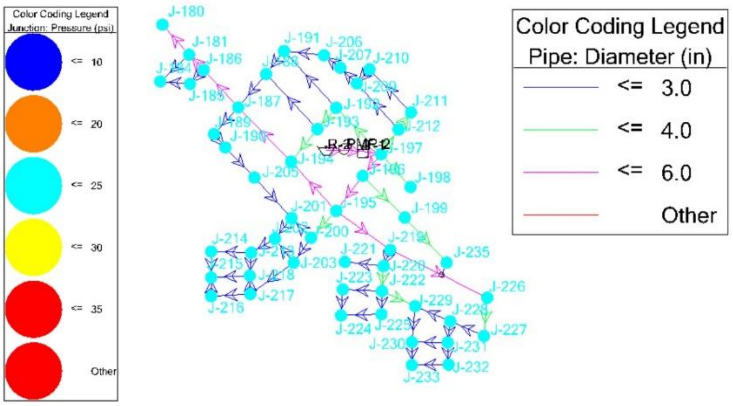


Figure 5. Complete Pumping with storage system

7.2 Nodes and Demand Allocation:

- Each residential area, mosque, and park was represented by a node.
- Demands at nodes shown in Annexure 1 were assigned based on:
    - Residential areas: Proportional to the number of houses.
    - Mosques, parks, and schools: Based on specific requirements.

7.3 Pipes:

- Hazen-Williams coefficient (C=130) was assigned for all GI pipes.
- Pipe diameters were trial-assigned and adjusted during the simulation to meet pressure and velocity criteria.

7.4 Pump and Reservoir

7.4.1 For Direct Pumping:

The pump was modeled with the required head-discharge curve, ensuring sufficient head for all nodes.

7.4.2 For Pumping with OHR:

The OHR was modeled with the specified height (**62 ft**) and capacity (**39 m³**).

7.5 Simulation Results (WaterGEMS Output)

7.5.1 Hydraulic Head at Nodes:

- The simulation ensured that all nodes maintained a **minimum residual pressure of 40 ft (12 m)** during peak demand.
- Nodes near the pump or OHR experienced higher pressures, which were reduced using pressure-reducing valves.

7.5.2 Pipe Velocities:

- Velocities in all pipes were maintained between **1.5–7 ft/sec** for the rising main and **1–5 ft/sec** for distribution pipes.

7.5.3 Energy Requirements:

- Direct Pumping: Higher energy demand due to continuous operation of the pump.
- OHR System: Lower operational energy requirements as water is distributed via gravity.

8 Pipe Summaries (Using WaterGEMS Reports)

WaterGEMS was used to extract pipe summaries, including diameters and lengths, for cost estimation. The output data is summarized below:

Table 1. Pressure pipe inventory Table

Pressure Pipes Inventory

Diameter (in)	Length (Ductile Iron) (ft)	Length (All Materials) (ft)	Volume (MG)
3.0	11,117	11,117	0.00
4.0	3,558	3,558	0.00
6.0	4,079	4,079	0.01
All Diameters	18,754	18,754	0.01

- These lengths will be used for material costing, as per the Market Rate Schedule (MRS).

8.1 Valve Placement

Valves were placed strategically in the model:

1. **Sluice Valves**: Installed at branch points to isolate sections during maintenance.
2. **Air Release Valves**: Placed at high points in the pipeline to release trapped air.
3. **Pressure-Reducing Valves**: Used to mitigate excessive pressures in high-pressure zones.

8.2 Attachments from WaterGEMS

The following simulation results from WaterGEMS are attached as annexures:

1. **Pressure Distribution Map**: Showing pressure at all nodes.
2. **Velocity Profiles**: Highlighting velocities in all pipes.
3. **Pipe and Node Table**: Summarizing node demands, pipe diameters, and lengths.

9 Cost Invloved

9.1 Pipe Costs

- From the **Pressure Pipes Inventory shown in Figure** :
    - **3-inch Pipes**: 11,117 ft × PKR 200/ft = PKR 2,223,400
    - **4-inch Pipes**: 3,558 ft × PKR 300/ft = PKR 1,067,400
    - **6-inch Pipes**: 4,079 ft × PKR 450/ft = PKR 1,835,550
    - **Total Pipe Cost**: **PKR 5,126,350**

9.2 Valve Costs

9.2.1 Sluice Valves:

Installed at **junctions** and at intervals of **1,000 ft** along straight pipes.

Total Sluice Valves = **51 (junctions)** + 18,754/1,000 = 19 = **70 valves**.

Total Cost = 70 valves × PKR 15,000/valve = **PKR 1,050,000**

9.2.2 Air Release Valves:

Installed at high points or intervals of 1,000 ft.

Total Air Release Valves = 18,754/1,000 = 19.

Total Cost = 19 valves × PKR 10,000/valve = **PKR 190,000**

9.2.3 Check Valves:

Installed near pumps.

Total Check Valves = **2** (1 per pump).

Total Cost = 2 valves × PKR 12,000/valve = **PKR 24,000**

Total Valve Cost: **PKR 1,264,000**

9.3 Excavation

- Excavation is for a total trench length of 18,754 ft. Assuming PKR 500 per cubic meter:

Total Cost = **PKR 500,000** (approx., based on assumptions).

9.4 Pumping System

- **Direct Pumping (10 HP Pump)**: **PKR 100,000**
- **Pumping with OHR (15 HP Pump): PKR 150,000**

9.5 Overhead Reservoir

- Capacity = 39 m^3 × PKR 20,000/m^3 = **PKR 780,000**

Table 2. Total Cost of all Components

Component	Quantity	Unit Cost (PKR) (Reference)	Total Cost (PKR)
3-inch Pipes	11,117	200/ft (MRS Lahore)	2,223,400
4-inch Pipes	3,558 ft	300/ft (MRS Lahore)	1,067,400
6-inch Pipes	4,079 ft	450/ft (MRS Lahore)	1,835,550
Sluice Valves	70 valves	15,000/valve (PHED Design Criteria)	1,050,000
Air Release Valves	19 valves	10,000/valve (PHED Design Criteria)	190,000
Check Valves	2 valves	12,000/valve (MRS Lahore)	24,000
Excavation	18,754 ft	500/cubic meter (MRS Lahore)	500,000 (approx.)
Pump (Direct)	1 × 10 HP	100,000 (MRS Lahore)	100,000
Pump (OHR)	1 × 15 HP	150,000 (MRS Lahore)	150,000
Overhead Reservoir	39 m³	20,000/m³ (MRS Lahore)	780,000

9.6 Total Capital Costs:

- Direct Pumping: **PKR 7,889,350**.
- Pumping with OHR: **PKR 9,597,950**.

9.7 Electricity Costs

Formula:

$$Cost\ (PKR/month) = HP \times Hours/month \times Electricity\ Rate \times 0.746$$

- **Direct Pumping** (10 HP):

$$10 \times 240 \times 20 \times 0.746 = \mathbf{35,856\ PKR/month}$$

- **Pumping with OHR** (15 HP):

$$15 \times 240 \times 20 \times 0.746 = \mathbf{53,784\ PKR/month}$$

9.8 Maintenance Costs

Maintenance Cost = 5% of Capital Cost per Year:

- **Direct Pumping**:

$$5\% \times 7{,}889{,}350 = 394{,}467.5\, PKR/year$$

- **Pumping with OHR**:

$$5\% \times 9{,}597{,}950 = 479{,}897.5\, PKR/year$$

Table 3. Total Maintenance cost

Component	Direct Pumping (PKR)	Pumping with OHR (PKR)
Electricity (Monthly)	35,856	53,784
Maintenance (Annual)	394,467.5	479,897.5

9.9 Total Operational Costs (20 Years):

- Direct Pumping: **PKR 8,605,440**.
- Pumping with OHR: **PKR 12,908,160**.

9.10 Total Cost Over 20 Years

- **Direct Pumping**:

$$Capital\ Cost + (20 \times Operational\ Cost)$$

=7,889,350 + (20×500,268) = 16,894,710 PKR

Pumping with OHR:

$$Capital\ Cost + (20 \times Operational\ Cost)$$

$$=9{,}597{,}950 + (20 \times 759{,}408) = 24{,}785{,}110\, PKR$$

Table 4. Total cost over 20 years

System	CapitalCost (PKR)	Operational Cost (20 Years)	Total Cost (PKR)
Direct Pumping	7,889,350	8,605,440	**16,494,790**
Pumping with OHR	9,597,950	12,908,160	$\mathbf{24,785,110}$

10 Staffing Levels and Costs

10.1 Staffing Levels

The operation and maintenance of the water supply system require the following staff:

Table 5. Staff levels

Role	Number of Staff	Responsibilities
System Supervisor	1	Overall management, compliance with PHED guidelines, scheduling, and reporting.
Technicians	4	Inspection, maintenance, and minor repairs of pumps, pipes, and valves.
Pump Operators	2	Operation, monitoring, and maintenance of pumps and reservoirs.
Laborers	6	Cleaning, flushing, and excavation as required for pipe and valve maintenance.
Administrative Staff	2	Documentation, inventory management, and record-keeping.

Total Staff Required: 15 personnel.

10.2 Cost of Staffing

The monthly and annual salary costs for each staffing role are outlined below:

Table 6. Staff Costing

Role	Monthly Salary (PKR)	Number of Staff	Total Monthly Cost (PKR)	Annual Cost (PKR)
System Supervisor	50,000	1	50,000	600,000
Technicians	40,000	4	160,000	1,920,000
Pump Operators	35,000	2	70,000	840,000
Laborers	25,000	6	150,000	1,800,000
Administrative Staff	30,000	2	60,000	720,000

Total Monthly Staffing Cost: PKR 490,000
Total Annual Staffing Cost: PKR 5,880,000

10.3 Justification for Staffing

- **System Supervisor** ensures efficient operation and regulatory compliance.
- **Technicians** and **Pump Operators** handle technical and operational aspects to ensure uninterrupted water supply.
- **Laborers** assist in manual tasks such as flushing, cleaning, and minor repairs.
- **Administrative Staff** manages records, inventory, and billing processes.

11 Final Comparison and Recommendation for Muslim Nagar Society

11.1 Direct Pumping:

Total Cost Over 20 Years: **PKR 16,894,710**

Pros:

- Lower initial capital cost.
- Lower operational costs due to smaller pump size.
- Simpler maintenance with fewer components.

Cons:

- Entirely dependent on electricity; any outages can disrupt supply.
- Pressure variations may occur in the network during peak demand.

11.2 Pumping with OHR:

Total Cost Over 20 Years: **PKR 24,785,110**

Pros:

- More reliable supply due to gravity-based distribution.
- Better pressure stability across the network.
- Suitable for areas with power outages.

Cons:

- Higher initial capital cost due to OHR construction and larger pump.
- Higher operational costs from increased energy consumption.

11.3 Recommendation for Muslim Nagar Society:

- Considering the stable power supply in **Lahore**, the **Direct Pumping System** is more cost-effective and practical for **Muslim Nagar Society**, especially given its lower total cost over 20 years.
- If future power outages are anticipated or additional reliability is required, the **Pumping with OHR System** may be reconsidered.

12 Monthly Household Bills for Operational Costs

For managing the total operational cost, the cost is divided among the houses and taken as monthly bills from the residets. To estimate the **monthly household bills** for managing the operational costs of each system, the following approach is used:

12.1 Total Operational Cost (Monthly):

- Direct Pumping: **PKR 41,689/month** (electricity + maintenance).
- Pumping with OHR: **PKR 63,284/month** (electricity + maintenance).

Total Number of Houses:
The society consists of **350 constructed houses.**

1. Household Contribution:
Monthly household bills are calculated by dividing the total operational cost by the total number of houses:

$$Monthly\ Bill\ per\ House = Total\ Monthly\ Operational\ Cost / Number\ of\ Houses$$

12.2 Monthly Bills

1. **Direct Pumping**:

$$Monthly\ Bill\ per\ House = \frac{41{,}689}{350} \approx 139\ PKR/house$$

2. **Pumping with OHR**:

$$Monthly\ Bill\ per\ House = \frac{63{,}284}{350} \approx 211\ PKR/house$$

12.3 Summary:

Table 7. Total Cost Summary

System	Total Monthly Cost (PKR)	Monthly Bill per House (PKR)
Direct Pumping	41,689	139
Pumping with OHR	63,284	211

12.4 Key Observations

- **Direct Pumping**: The monthly household bill is **PKR 139**, which is affordable and sustainable for the residents.
- **Pumping with OHR**: The monthly household bill is **PKR 211**, which is higher but may be justified for the added reliability and pressure stability.

12.5 Recommendation:

For **Muslim Nagar Housing Society**, **Direct Pumping** provides a more economical solution for residents, with significantly lower household bills. If the residents prioritize reliability and are willing to pay higher bills, the **Pumping with OHR** system may be an option.

13 References

1. *Water and Sanitation Agency (WASA) Design Guidelines: Standards for water supply systems, including pressure and velocity criteria.*
2. *Public Health Engineering Department (PHED) Punjab: Criteria for water consumption, pipe sizing, and fire safety.*
3. *Market Rate Schedule (MRS) Lahore: Pricing for pipes, valves, and construction materials.*
4. *WaterGEMS/EPANET: Hydraulic modeling software used for designing and simulating the water supply system.*
5. *Population Growth Rate of Lahore: Statistics used to project population over 20 years (2.5% annual growth).*
6. *Google Earth: Reference for geographic and spatial data of the project area.*
7. *Society Head Office: Source of information regarding residential and community facilities in Muslim Nagar Housing Society.*
8. *Hazen-Williams Equation: Formula used for calculating head losses in pipes.*
9. *Government of Punjab: Guidelines for overhead reservoir sizing and operational criteria.*
10. *Literature on Climatic Data: Sources for Lahore's temperature, rainfall, and weather patterns.*

14 Annexure

JUNCTIONS DEMAND

ID	Label	Unit Dem	Demand (gpm)
467	J-180	<Collecti	8
468	J-181	<Collecti	6
473	J-184	<Collecti	6
475	J-185	<Collecti	2
477	J-186	<Collecti	2
480	J-187	<Collecti	9
482	J-188	<Collecti	9
484	J-189	<Collecti	7
486	J-190	<Collecti	7
488	J-191	<Collecti	7
490	J-192	<Collecti	15
492	J-193	<Collecti	8
495	J-194	<Collecti	13
498	J-195	<Collecti	16
500	J-196	<Collecti	9
502	J-197	<Collecti	11
505	J-198	<Collecti	9
507	J-199	<Collecti	6
509	J-200	<Collecti	9
511	J-201	<Collecti	19
513	J-202	<Collecti	4
515	J-203	<Collecti	6
519	J-205	<Collecti	6
522	J-206	<Collecti	11
524	J-207	<Collecti	9
528	J-209	<Collecti	6
530	J-210	<Collecti	10
532	J-211	<Collecti	12
534	J-212	<Collecti	13
538	J-213	<Collecti	11
540	J-214	<Collecti	3
542	J-215	<Collecti	6
544	J-216	<Collecti	3
546	J-217	<Collecti	12
548	J-218	<Collecti	6
554	J-219	<Collecti	7
556	J-220	<Collecti	9
558	J-221	<Collecti	5
560	J-222	<Collecti	16
562	J-223	<Collecti	11
564	J-224	<Collecti	11
566	J-225	<Collecti	6
569	J-226	<Collecti	8
571	J-227	<Collecti	10
573	J-228	<Collecti	4

575	J-229	<Collecti	9
578	J-230	<Collecti	6
580	J-231	<Collecti	9
583	J-232	<Collecti	10
585	J-233	<Collecti	9
590	J-235	<Collecti	3

PIPE LENGTH AND DIAMETER

ID	Label	Length (S	Start	Stop Node	Diamete	Mater	Hazen-W	Flow (gp	Velocity	Headloss Gradient
469	P-166	293	J-180	J-181	6	Ductile	130	-8	0.09	0
474	P-168	286	J-181	J-184	3	Ductile	130	3	0.15	0
476	P-169	217	J-184	J-185	3	Ductile	130	-2	0.1	0
478	P-170	144	J-185	J-186	3	Ductile	130	-5	0.21	0
479	P-171	153	J-186	J-181	6	Ductile	130	17	0.19	0
481	P-172	373	J-186	J-187	6	Ductile	130	-24	0.27	0
483	P-173	318	J-187	J-188	3	Ductile	130	-15	0.68	0.001
485	P-174	263	J-187	J-189	3	Ductile	130	29	1.33	0.003
487	P-175	127	J-189	J-190	3	Ductile	130	22	1.01	0.002
489	P-176	213	J-188	J-191	3	Ductile	130	-15	0.68	0.001
491	P-177	562	J-191	J-192	3	Ductile	130	-14	0.62	0.001
493	P-178	207	J-192	J-193	4	Ductile	130	51	1.3	0.002
494	P-179	549	J-193	J-188	3	Ductile	130	9	0.39	0
496	P-180	305	J-193	J-194	4	Ductile	130	34	0.88	0.001
497	P-181	553	J-194	J-187	6	Ductile	130	47	0.53	0
499	P-182	487	J-194	J-195	6	Ductile	130	-26	0.29	0
501	P-183	321	J-195	J-196	6	ctile Ir	130	-243	2.76	0.005
503	P-184	207	J-196	J-197	6	ctile Ir	130	-261	2.96	0.006
504	P-185	471	J-197	J-192	4	Ductile	130	79	2.02	0.005
506	P-186	322	J-197	J-198	4	Ductile	130	9	0.24	0
508	P-187	432	J-196	J-199	4	Ductile	130	9	0.24	0
510	P-188	267	J-195	J-200	4	Ductile	130	71	1.81	0.004
512	P-189	202	J-200	J-201	3	Ductile	130	29	1.32	0.003
514	P-190	207	J-201	J-202	3	ctile Ir	130	20	0.9	0.002
516	P-191	209	J-202	J-203	3	ctile Ir	130	-8	0.38	0
520	P-193	306	J-190	J-205	3	Ductile	130	15	0.69	0.001
521	P-194	401	J-205	J-201	3	Ductile	130	10	0.44	0
523	P-195	294	J-191	J-206	3	Ductile	130	-9	0.39	0
525	P-196	150	J-206	J-207	3	Ductile	130	-20	0.9	0.002
529	P-198	168	J-207	J-209	3	Ductile	130	-29	1.33	0.003
531	P-199	138	J-209	J-210	3	Ductile	130	-11	0.51	0.001
533	P-200	437	J-210	J-211	3	Ductile	130	-22	0.98	0.002
535	P-201	155	J-211	J-212	4	Ductile	130	-33	0.85	0.001
536	P-202	453	J-212	J-209	3	Ductile	130	24	1.1	0.002
537	P-203	220	J-212	J-197	4	Ductile	130	-70	1.8	0.004
539	P-204	193	J-202	J-213	3	Ductile	130	24	1.09	0.002
541	P-205	292	J-213	J-214	3	Ductile	130	6	0.26	0

543	P-206	189	J-214	J-215	3	Ductile	130	3	0.12	0
545	P-207	137	J-215	J-216	3	Ductile	130	-1	0.02	0
547	P-208	287	J-216	J-217	3	Ductile	130	-4	0.17	0
549	P-209	134	J-217	J-218	3	Ductile	130	3	0.12	0
550	P-210	168	J-218	J-213	3	Ductile	130	-7	0.32	0
551	P-211	283	J-218	J-215	3	Ductile	130	3	0.15	0
552	P-212	390	J-217	J-203	3	Ductile	130	-18	0.83	0.001
553	P-213	223	J-203	J-200	3	Ductile	130	-33	1.5	0.004
555	P-214	487	J-195	J-219	6	Ductile	130	130	1.48	0.002
557	P-215	128	J-219	J-220	3	Ductile	130	67	3.06	0.015
559	P-216	285	J-220	J-221	3	Ductile	130	5	0.21	0
561	P-217	183	J-220	J-222	4	Ductile	130	53	1.36	0.002
563	P-218	286	J-222	J-223	3	Ductile	130	14	0.63	0.001
565	P-219	189	J-223	J-224	3	Ductile	130	3	0.13	0
567	P-220	295	J-224	J-225	3	Ductile	130	-8	0.37	0
568	P-221	167	J-225	J-222	3	Ductile	130	-15	0.66	0.001
570	P-222	791	J-219	J-226	6	Ductile	130	56	0.64	0
572	P-223	278	J-226	J-227	4	Ductile	130	48	1.23	0.002
574	P-224	269	J-227	J-228	3	Ductile	130	38	1.72	0.005
576	P-225	278	J-228	J-229	3	Ductile	130	12	0.52	0.001
577	P-226	267	J-229	J-222	4	Ductile	130	-9	0.23	0
579	P-227	263	J-229	J-230	3	Ductile	130	12	0.54	0.001
581	P-228	262	J-230	J-231	3	Ductile	130	-3	0.14	0
582	P-229	156	J-231	J-228	3	Ductile	130	-22	1.01	0.002
584	P-230	161	J-231	J-232	3	Ductile	130	10	0.47	0
586	P-231	268	J-232	J-233	3	Ductile	130	0	0.01	0
587	P-232	167	J-233	J-230	3	Ductile	130	-9	0.42	0
591	P-234	450	J-199	J-235	4	Ductile	130	3	0.08	0
603	P-235	135	T-1	J-197	6	Ductile	130	431	4.89	0.016
621	P-242	132	PMP-	T-1	6	Ductile	130	539	6.12	0.024
622	P-243	147	PMP-	R-2	6	Ductile	130	-539	6.12	0.024

Pump Definition

ID	Label	Pump De	Statu	Hydraulic	Hydraul	Flow (	Pump Head (ft)
619	PMP-2	Pump De	On	-3.47	58.39	539	61.86

TANK

ID	Label	Elevation	Diam	Flow (Out	Hydraulic Grade (ft)
602	T-1	52	10	-108	55.28

RESERVIOR

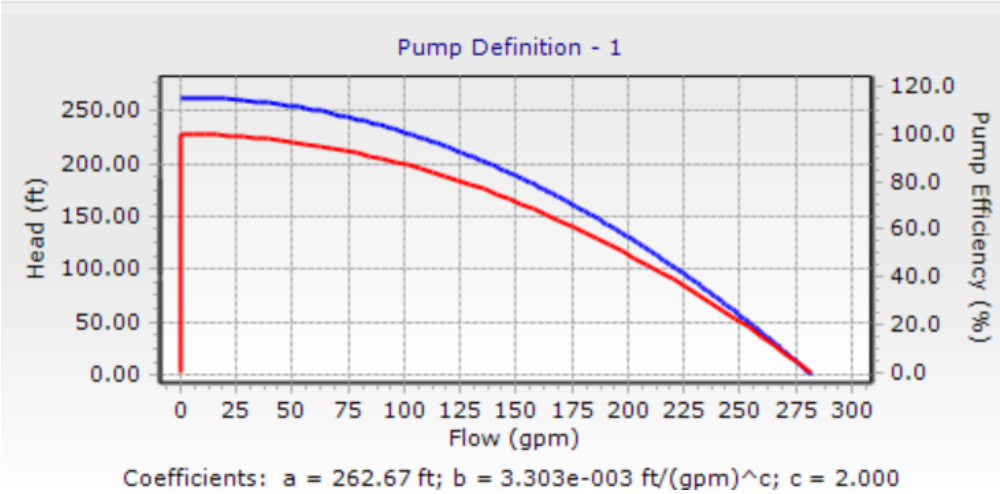
ID	Label	Flow (Ou	Hydraulic Grade (ft)
620	R-2	539	0

1. **Screen shots of the pump definition curve.**

A. PUMPING WITH STORAGE

Pump Definition Type: Design Point (1 Point)

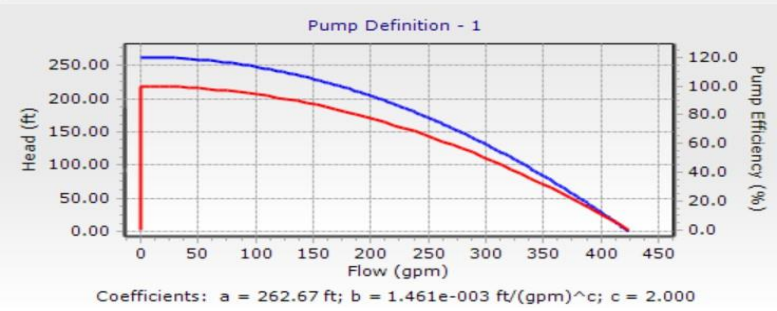
	Flow (gpm)	Head (ft)
Shutoff:	0	262.67
Design:	141	197.00
Max. Operating:	282	0.00



B. DIRECT PUMPING

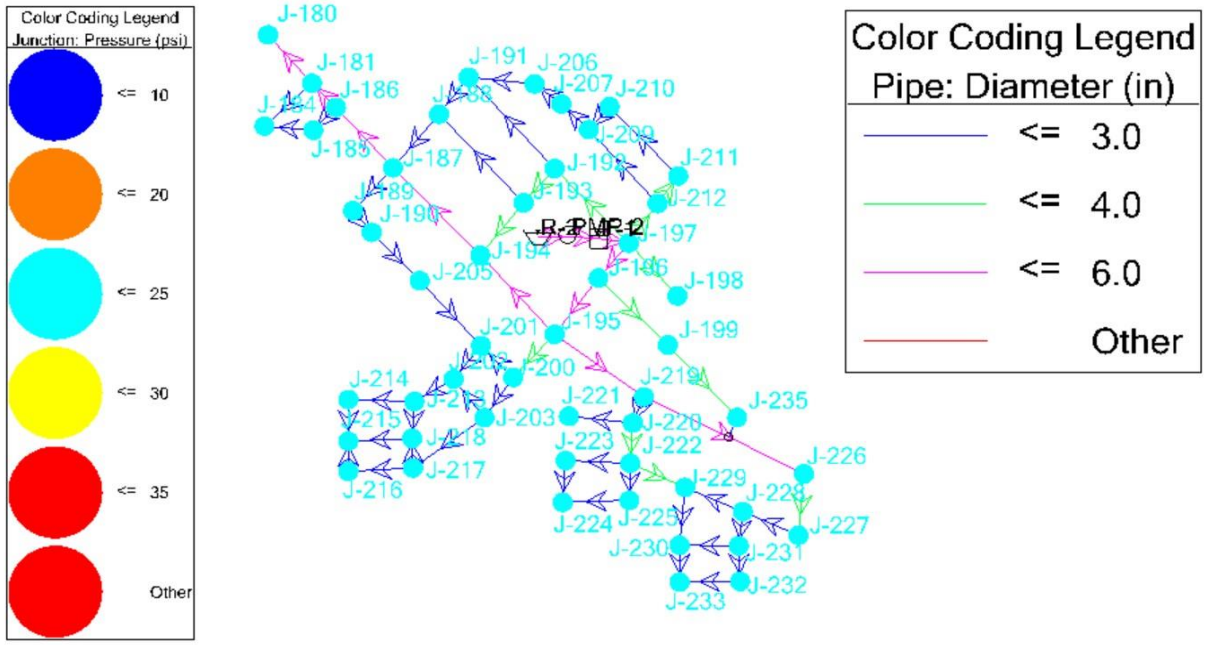
Pump Definition Type: Design Point (1 Point)

	Flow (gpm)	Head (ft)
Shutoff:	0	262.67
Design:	212	197.00
Max. Operating:	424	0.00

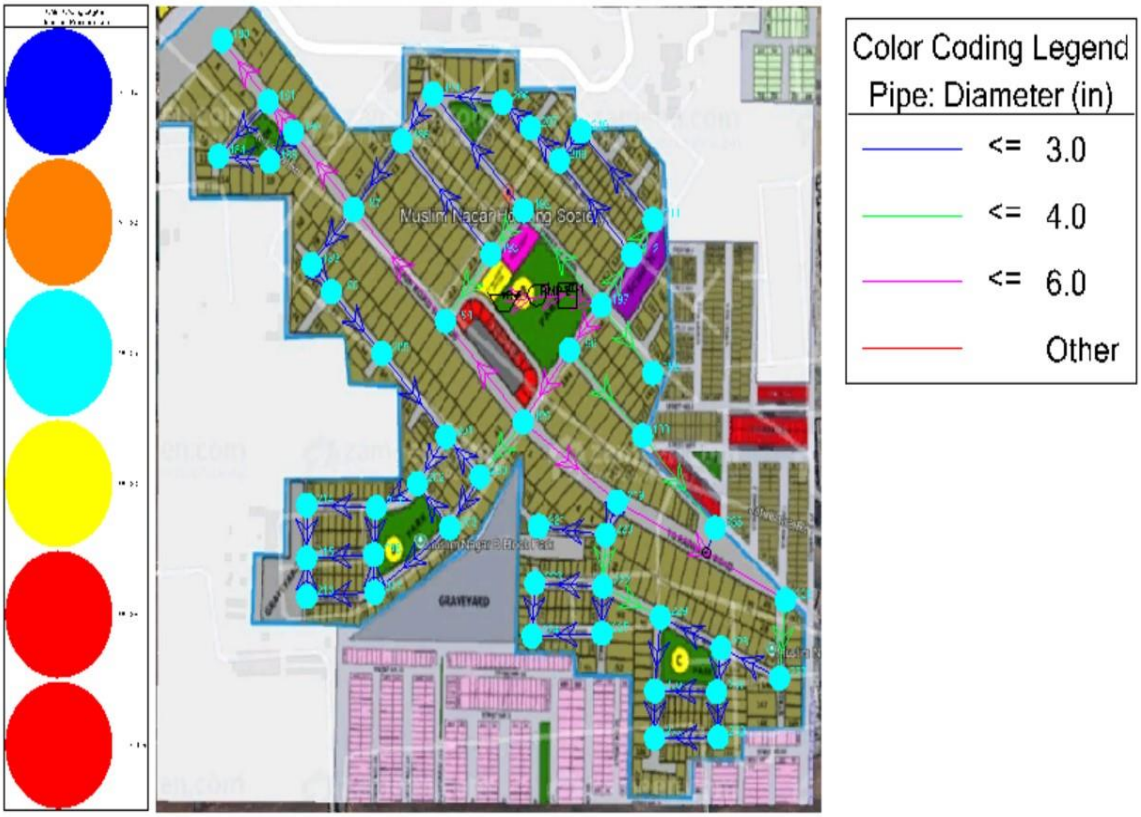


Screen shot of the distribution layout plan along with backdrop image showing nodes and pipes

(ft/ft)



1. **Screen shot of the distribution layout plan with color coding of diameters and pressures.**



Pressure Pipes Inventory

Pressure Pipes Inventory

Diameter (in)	Length (Ductile Iron) (ft)	Length (All Materials) (ft)	Volume (MG)
3.0	11,117	11,117	0.00
4.0	3,558	3,558	0.00
6.0	4,079	4,079	0.01
All Diameters	18,754	18,754	0.01

Lateral Inventory

	Count	Total Length (ft)
All Laterals	1	96

Hydraulic Model Inventory: Untitled1.wtg

Title	
Engineer	
Company	
Date	20-Sep-24
Notes	

Scenario Summary

ID	1
Label	Base
Notes	
Active Topology	Base Active Topology
Physical	Base Physical
Demand	Base Demand
Initial Settings	Base Initial Settings
Operational	Base Operational
Age	Base Age
Constituent	Base Constituent
Trace	Base Trace
Fire Flow	Base Fire Flow
Energy Cost	Base Energy Cost
Transient	Base Transient
Pressure Dependent Demand	Base Pressure Dependent Demand
Failure History	Base Failure History
SCADA	Base SCADA
User Data Extensions	Base User Data Extensions
Steady State/EPS Solver Calculation Options	Base Calculation Options
Transient Solver Calculation Options	Base Calculation Options

Network Inventory

Pipes	68	-Standard (3 Point)	0
Laterals	1	-Standard Extended	0
Junctions	51	-Constant Speed - Four-Quadrant Characteristics	1
Hydrants	0	-Constant Speed - Pump Definition	0
Tanks	1	-Shut Down After Time Delay	0
-Circular	1	-Variable Speed/Torque	0
-Non-Circular	0	-Pump Start - Variable Speed/Torque	0
-Variable Area	0	Pump Stations	0
Reservoirs	1	Variable Speed Pump Batteries	0
Customer Meters	0	PRV's	0
Taps	1	PSV's	0
SCADA Elements	0	PBV's	0
Pumps	1	FCV's	0
-Constant Power	0	TCV's	0
-Custom Extended	0	GPV's	0
-Design Point (1 Point)	1	Isolation Valves	0
-Multiple Point	0	Spot Elevations	0

Hydraulic Model Inventory: Untitled1.wtg

Transient Network Inventory			
Turbines	0	Rupture Disks	0
Periodic Head-Flows	0	Discharges to Atmosphere	0
Air Valves	0	Orifices Between Pipes	0
Hydropneumatic Tanks	0	Valves With Linear Area Change	0
Surge Valves	0	Surge Tanks	0
Check Valves	0		

Pressure Pipes Inventory			
3.0 (in)	11,117 ft	6.0 (in)	4,079 ft
4.0 (in)	3,558 ft	All Diameters	18,754 ft

Lateral Inventory			
Count	1	Total Length	96 ft

Scenario Summary Report
Scenario: Base

Scenario Summary

ID	1
Label	Base
Notes	
Active Topology	Base Active Topology
Physical	Base Physical
Demand	Base Demand
Initial Settings	Base Initial Settings
Operational	Base Operational
Age	Base Age
Constituent	Base Constituent
Trace	Base Trace
Fire Flow	Base Fire Flow
Energy Cost	Base Energy Cost
Transient	Base Transient
Pressure Dependent Demand	Base Pressure Dependent Demand
Failure History	Base Failure History
SCADA	Base SCADA
User Data Extensions	Base User Data Extensions
Steady State/EPS Solver Calculation Options	Base Calculation Options
Transient Solver Calculation Options	Base Calculation Options

Hydraulic Summary

Time Analysis Type	Steady State	Use simple controls during steady state?	True
Friction Method	Hazen-Williams	Is EPS Snapshot?	False
Accuracy	0.001	Start Time	12:00:00 AM
Trials	40	Calculation Type	Hydraulics Only